

Sinkhorn scheme for forward-starter options

If instead $X = S_{T_2}$ and $Y = S_{T_1}$ for a martingale stock price process S with $0 < T_1 < T_2$ and μ_Y and μ_X are in **convex order** (see FM02 for definition), then our third marginal constraint $\int_{[0,\infty) \times [0,\infty)} (x - Ky)^+ \mu(dx, dy) = \int_{[0,\infty)} (z - K)^+ \mu_3(dz)$ for all $K \geq 0$ now corresponds to having observed option prices for **forward-starter** options for all strikes rather than cross-rate options as above. In this case, we require the additional martingale condition $\mathbb{E}(X|Y) = Y$.

At this point we make the following natural assumption:

Assumption 0.1 *There exists a $\mu^* \in \mathcal{P}(\mu_X, \mu_Y, \mu_Z)$ with $H(\mu^*|\bar{\mu}) < \infty$ which satisfies the martingale condition above.*

Then (assuming e.g. that $\bar{\mu}(x, y) = \mu_X(x) \frac{1}{x\sigma_2\sqrt{2\pi}} e^{-\frac{(\log \frac{x}{y} + \frac{1}{2}\sigma_2^2)^2}{2\sigma_2^2}}$ for some σ_2 so as to respect the martingale condition) the martingale condition $\mathbb{E}(X - Y|Y) = 0$ leads to a fourth Sinkhorn equation:

$$0 = \int_0^\infty (x - y) e^{u_{n+1}(x) + y w_{n+1}(\frac{x}{y}) + \Delta_{n+1}(y)(x-y)} \frac{\bar{\mu}(x, y)}{\mu_Y(y)} dx \quad (1)$$

for Δ_{n+1} . Note we have divided out the $e^{v_{n+1}(y)}$ term here since we have zero on the right hand side and this terms do not depend on the variable of integration x , and the other three Sinkhorn equations take a similar form as before but with an additional hedging term in the exponent:

$$\begin{aligned} u_{n+1}(x) &:= \log \mu_X(x) - \log \int_0^\infty e^{v_n(y) + y w_n(\frac{x}{y}) + \Delta_n(y)(x-y)} \bar{\mu}(x, y) dy \\ v_{n+1}(y) &:= \log \mu_Y(y) - \log \int_0^\infty e^{u_{n+1}(x) + y w_{n+1}(\frac{x}{y}) + \Delta_n(y)(x-y)} \bar{\mu}(x, y) dx \\ \mu_Z(z) &= \int_0^\infty e^{u_{n+1}(x) + v_{n+1}(\frac{x}{z}) + \frac{x}{z} w_{n+1}(z) + \Delta_n(\frac{x}{z})(x - \frac{x}{z})} \frac{x^2}{z^3} \bar{\mu}(x, \frac{x}{z}) dx. \end{aligned} \quad (2)$$

For each fixed y , Eq (1) is just a one-dimensional root-finding exercise for $\Delta = \Delta(y)$.